JAN-T-602

26 APRIL 1948

JOINT ARMY-NAVY SPECIFICATION

TITANIUM DIOXIDE (OINTMENT TYPE)

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments, and supersedes the following specification:

U. S. Army
97–54–353
14 Sept. 1945

A. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATIONS

A–1. *Specifications.*—The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

JOINT ARMY-NAVY SPECIFICATION
JAN–P–118—Packaging and Packing for Overseas Shipment—Sacks, Paper, Shipping, Multiwall.

U. S. ARMY SPECIFICATION
100–2—Standard Specification for Marking Shipments by Contractors.[1]

NAVY DEPARTMENT SPECIFICATION
General Specification for Inspection of Material.[2]

FEDERAL SPECIFICATION
RR–S–366—Sieves; Standard, Testing.

A–2. *Other publications.*—The following publications, of the issue in effect on date of invitation for bids, form a part of this specification:

BUREAU OF SUPPLIES AND ACCOUNTS PUBLICATION
Navy Shipment Marking Handbook.[2]

INTERSTATE COMMERCE COMMISSION REGULATIONS
Regulations for Transportation of Explosives and Other Dangerous Articles, etc.

B. GRADE

B–1. This specification covers one grade of titanium dioxide as hereinafter specified. (See par. H–1.)

C. MATERIAL AND WORKMANSHIP

C–1. *Material.*—The material shall consist of a finely divided titanium dioxide pigment of the anatase crystal form.

D. GENERAL REQUIREMENTS

D–1. See section E.

[1] Applicable only to Army purchases.
[2] Applicable only to Navy purchases.

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E. DETAIL REQUIREMENTS

E–1. The material shall comply with the detail requirements listed in table I.

TABLE I.—*Detail requirements.*

	Minimum	Maximum
Titanium dioxide (percentage)	98.0	...
Anatase crystal form (percentage)	90.0	...
Retained on No. 325 U. S. Standard sieve[1]	....	0.1
Moisture (percentage)	....	0.3
Trace elements (calculated as metal):		
Lead[2] (ppm.)	....	200
Iron[2] (ppm.)	....	100
Copper[2] (ppm.)	....	15
Arsenic[2] (ppm.)	....	10

[1] Sieves shall conform to Federal Specification RR-S-366.
[2] The manufacturer shall, at the time of forwarding his bid, submit for approval by the procuring agency, a complete description of the methods of analysis he proposes to use in the certification required in paragraph F-4e.

F. METHODS OF SAMPLING, INSPECTION, AND TESTS

F–1. The material and work shall be subject at all times to inspection at the place of manufacture and every facility shall be afforded authorized Government inspectors for the prosecution of their work.

F–2. *Size of lot.*—Unless otherwise specified in the contract or order, each shipment shall constitute a lot, but shall not exceed 100,000 pounds.

F–3. *Sampling.*—Five containers shall be chosen, at random, from each lot for sampling purposes. If there are fewer than five containers in the lot, sample each container. Withdraw a ½-pound representative sample from each container so chosen. Place the samples in a clean, dry container, and mix thoroughly to form a composite sample representative of the lot. Seal and label the container to insure the identity of the composite sample with the lot represented. (See par. H–5.)

F–4. *Tests.*—All tests shall be conducted in accordance with procedures stated herein. Analytical reagent grade chemicals and distilled water shall be used throughout the tests. Blank determinations shall be run in parallel with the tests, using the same quantities of reagents as used in the test, and corrections shall be applied when significant.

F–4a. *Purity.*—

F–4a(1). *Ferric sulfate solution.*—Dissolve 20 gm. of pure iron or plain carbon steel in a slight excess of concentrated hydrochloric acid, oxidize with concentrated nitric acid, add 80 ml.

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concentrated sulfuric acid and heat to fumes of SO_3. Cool, carefully dilute with water to 1000 ml., digest on a steam bath until sulfates are dissolved and filter if necessary. To oxidize any ferrous iron that may be present, add 0.1*N* potassium permanganate solution until a faint pink color persists for 5 minutes.

F–4a(2). *Procedure.*—Weigh 0.3000 to 0.3300 gm. of the sample into a 250-ml. beaker, add 20 ml. concentrated sulfuric acid and 7 to 8 gm. ammonium sulfate. Mix well, and heat on a hot plate to fumes of SO_3 and continue to heat over a strong flame until solution is complete (usually not more than 5 minutes boiling is required), or it is apparent that the residue is siliceous matter. Cool, carefully dilute with 100 ml. water, stir, heat carefully to boiling while stirring, let settle, filter through paper and transfer the residue completely to the paper. Wash the residue with cold diluted sulfuric acid (1:19) until titanium is removed. Dilute filtrate and washings to 200 ml. and add approximately 5 ml. concentrated ammonium hydroxide to lower the acidity to 10 to 15 percent sulfuric acid (by volume). Wash out a Jones reductor (see par. H–4) with sulfuric acid (1:19) and water, leaving sufficient water in the reductor to fill to the upper level of the zinc. (These washings should require not more than 1 or 2 drops of 0.1*N* potassium permanganate solution to obtain a pink color.) Empty the receiver and measure 25 ml. of ferric sulfate solution (see par. F–4a(1)) from a graduate into the receiver. Reduce the prepared titanium solution as follows:

(a) Run 50 ml. sulfuric acid (1:19) through the reductor at a uniform rate so as to require 5 to 10 minutes for passage.
(b) Follow this with the titanium solution at a uniform rate so as to require 10 minutes for passage.
(c) Wash out with 100 ml. sulfuric acid (1:19).
(d) Finally run through about 100 ml. water. Care should be taken that the reductor is always filled with solution or water to the upper level of the zinc. Gradually release the suction; wash thoroughly the glass tube that was immersed in the ferric sulfate solution.

Remove the receiver and titrate immediately with standard 0.1*N* potassium permanganate solution.

$$\text{Percentage } TiO_2 = \frac{(A \times N) \times 7.99}{W}$$

where:

A = ml. of standard potassium permanganate solution
N = normality of standard potassium permanganate solution
W = weight of sample.

F–4b. *Anatase crystal form.*—The Government inspector shall obtain from the manufacturer certification that the material meets the crystal form requirements. However, the contracting officer may, at his discretion, have X-ray diffraction determinations made to assure that the material meets the crystal form requirements.

F–4c. *Sieve size.*—Weigh approximately 100 gm. of sample into a beaker, add 1000 ml. of water and mix thoroughly. Pass the

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slurry through a No. 325 U. S. Standard sieve. Wash the residue with water brushing gently with a camels-hair brush to facilitate passage of small particles through the screen. Wash the residue with water, dry in an oven at 100° to 110°C., cool in a desiccator, transfer the residue to a tared watch glass and weigh. Calculate the percentage of material retained by the sieve.

F–4d. *Moisture.*—Weigh approximately 5.00 gm. of sample into a tared wide mouth weighing bottle. Dry in an oven for 2 hours at 100° to 110°C., cool in a desiccator and weigh. Calculate the loss in weight as percentage of moisture.

F–4e. *Trace elements.*—The Government inspector shall obtain from the manufacturer certification that the material complies with the requirements for trace elements, when tested in accordance with methods submitted to and approved by the procuring agency.

G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT

G–1. *Packing.*—Titanium dioxide shall be packed either in 50-pound fiber drums complying with the requirements of Specification 21A of Interstate Commerce Commission Regulations for Transportation of Explosives and Other Dangerous Articles, etc., or in 50-pound, type B, style 2–X, multiwall shipping sacks complying with the requirements of Joint Army-Navy Specification JAN–P–118, as specified in the contract or order. (See par. H–2.) After filling, the top closure shall be sewn in the same manner as the bottom closure.

G–2. *Marking.*—In addition to any special marking required by the contract or order, shipping containers for the Army shall be marked in accordance with the requirements of U. S. Army Specification 100–2; for the Navy in accordance with the requirements of the Navy Shipment Marking Handbook.

H. NOTES

H–1. *Use.*—Material covered by this specification is intended for use as a component of protective ointment.

H–2. Requests, requisitions, schedules, and contracts or orders should contain the following features:

(a) Title, number and date of the specification
(b) Type of packing desired. (See par. G–1.)

H–3. This specification is intended to cover a grade of titanium dioxide pigment similar or equal to any of the following commercial products:

Ti-Pure FF—E. I. duPont de Nemours and Co., Inc.
Titanox A-Mo—National Lead Co.
Zopacque SD—Glidden Co.
Unitane 0-220—American Cyanamide Co.

H–4. Directions for preparing a Jones reductor may be found in Blair, "The Chemical Analysis of Iron", eighth edition, pages 88-89, or Treadwell-Hall, "Analytical Chemistry", volume 2, fifth edition.

H–5. If the contractor consistently produces high-quality material and operates under a system of quality control acceptable to the Government, the Government, at its discretion and with the

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concurrence of the contractor, may modify, in whole or in part, the sampling and testing procedures specified in this specification. However, the Government reserves the right to return at any time without previous notice to the contractor, to the sampling and testing procedures prescribed in this specification.

H–6. Information as to the availability of Interstate Commerce Commission Regulations for Transportation of Explosives and Other Dangerous Articles, etc., may be obtained from the Interstate Commerce Commission, Washington 25, D. C.

H–7. Copies of Joint Army-Navy specifications and Federal specifications (required for Army purchases) and U. S. Army specifications may be obtained as indicated in the "Index of United States Army, Joint Army-Navy and Federal Specifications Used by the War Department". Copies of this index may be obtained from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Agencies within the Department of the Army will obtain copies of Joint Army-Navy, U. S. Army, and Federal specifications through established departmental channels. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

H–8. Copies of Joint Army-Navy specifications and Federal specifications (required for Navy purchases), Navy Department specifications, and the Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Center, Norfolk 11, Va. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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